Fiscal Year:	FY 2005	**Task Last Updated:**	FY 07/01/2005
PI Name:	Ronney, Paul D. Sc.D.		
Project Title:	Studies of Premixed Laminar and Turbulent Flames at Microgravity (NAG3-2124)		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous flames - premixed		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Los Angeles	**State:**	CA
Zip Code:	90089-1453	**Congressional District:**	33
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	OTHER
Start Date:	03/01/1991	**End Date:**	07/31/2005
No. of Post Docs:	3	**No. of PhD Degrees:**	4
No. of PhD Candidates:	4	**No. of Master' Degrees:**	3
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	17
No. of Bachelor's Candidates:	15	**Monitoring Center:**	NASA GRC
Contact Monitor:	Weiland, Karen	**Contact Phone:**	216-433-3623
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Flight Program:	Shuttle		
Flight Assignment:	Note: grant extended 1 year		
Key Personnel Changes/Previous PI:	4		
COI Name (Institution):			
Grant/Contract No.:	NAG3-2124/NAG3-2887		
Performance Goal No.:	1.22		
Performance Goal Text:	Enable increased combustion system efficiency, reduced pollution, and mitigation of fire risks through insights and data bases obtainable only through microgravity experiments.		

Task Description:	Several studies relating to premixed laminar and turbulent flames at microgravity are being pursued. The primary task is the space flight experiment entitled “Structure Of Flame Balls At Low Lewis-number (SOFBALL)” that is investigating spherical, stable, stationary flame structures (“flame balls”) at microgravity. In addition to the space experiment and supporting ground-based microgravity experiments on the KC-135 aircraft, numerical modeling of flame balls with detailed chemical, transport and radiation emission/absorption models is being conducted. Related projects include investigation of “flame tubes” that have been discovered in stretched counterflow flames at low Lewis number, edge flames, flame propagation in Hele-Shaw cells, and experimental simulation of premixed combustion using aqueous autocatalytic chemical reactions. The objective of this work is to obtain a better understanding of the mechanisms of flammability, instability and ignition limits of premixed-gas flames and the ways in which buoyancy affects these limits. The results of these studies will lead to an improved understanding of the fire hazards that may exist in orbiting spacecraft and of ways to minimize these hazards. In particular, the SOFBALL experiments provide the opportunity to study the interactions of the two most important processes necessary for combustion (chemical reaction and heat and mass transport) in the simplest possible configuration.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	These studies may lead to an improved understanding of the mechanism of combustion limits in Earth-based devices, which in turn could lead to the development of cleaner and more efficient engines through the use of lean premixed combustors.
Task Progress:	Analysis of results from STS-107 are nearing completion and are being integrated with results from STS-83 and STS-94 in preparation for the project final report.
Bibliography Type:	Description: (Last Updated: 01/29/2007)